

Modeling and simulation of hypersonic re-entry: theoretical background

F. Bonelli

Dipartimento di Meccanica Matematica e Management, Politecnico di Bari, Italy

✉ francesco.bonelli@poliba.it



Dipartimento
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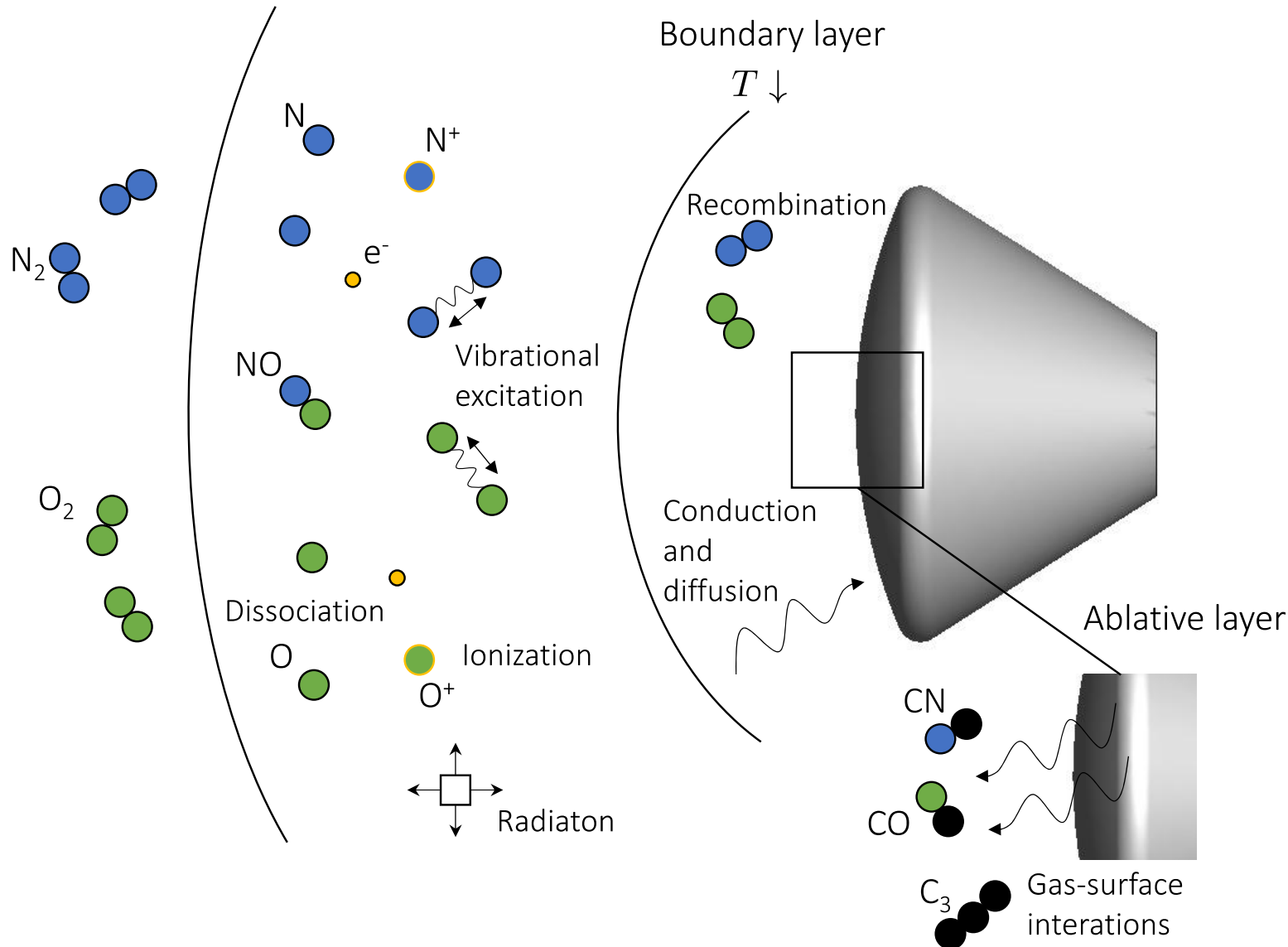
MUR
Dipartimento
di Eccellenza
2018-2022
2023-2027

Outline

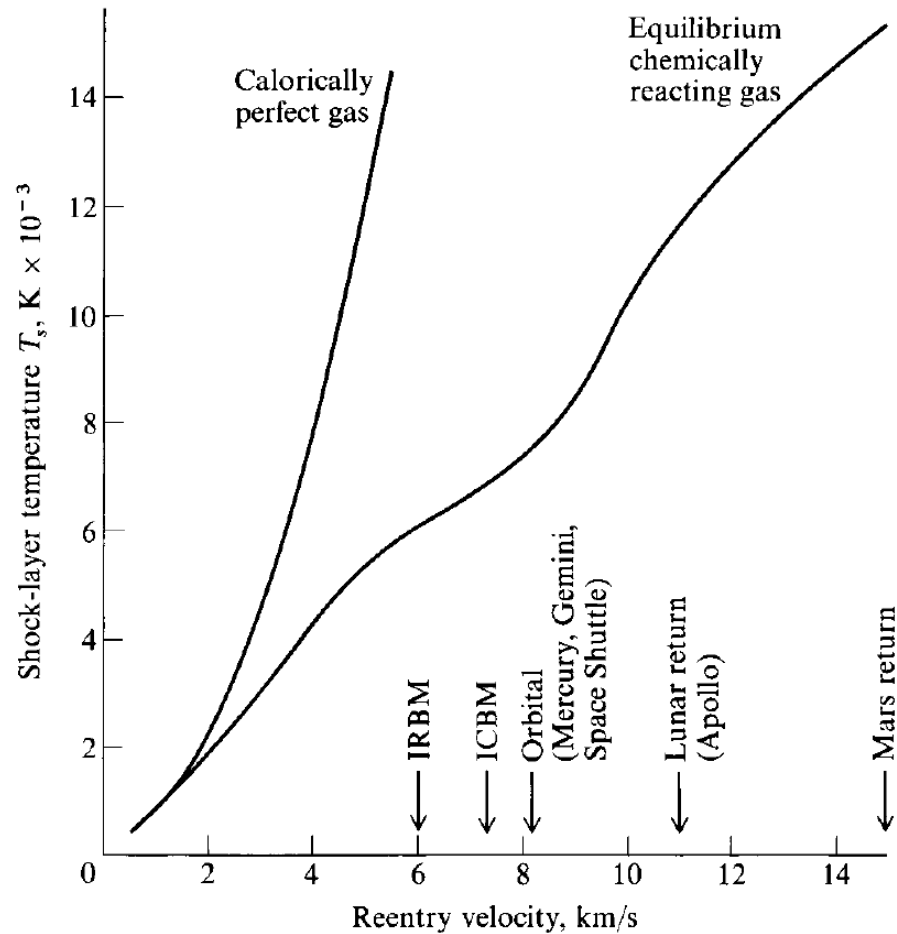
- **Hypersonic flows and high temperature effects**
- **Thermochemical non equilibrium models: neutral air mixture – State to State and Park's model**
- **Governing equations**
- **Numerical method**
- **Results for neutral air mixtures**
- **Multi-internal-temperature kinetic model (MiT): results and discussion**
- **Finite rate catalysis model: results and discussion**

Hypersonic flows: atmospheric entry scenario

Bow shock
 $T \approx 10000\text{ K}$



Calorically perfect gas vs local chemical equilibrium



52 km Altitude;

24 July 1969 Apollo 11 re-entry:

- Altitude 53 km
- $u=11.2$ km/s
- $T_\infty=283$ K
- Mach 32.5

Calorically perfect gas assumption

$$\frac{T_2}{T_1} = \left[1 + \frac{2\gamma}{\gamma + 1} (M_1^2 - 1) \right] \left[\frac{2 + (\gamma - 1)M_1^2}{(\gamma + 1)M_1^2} \right]$$

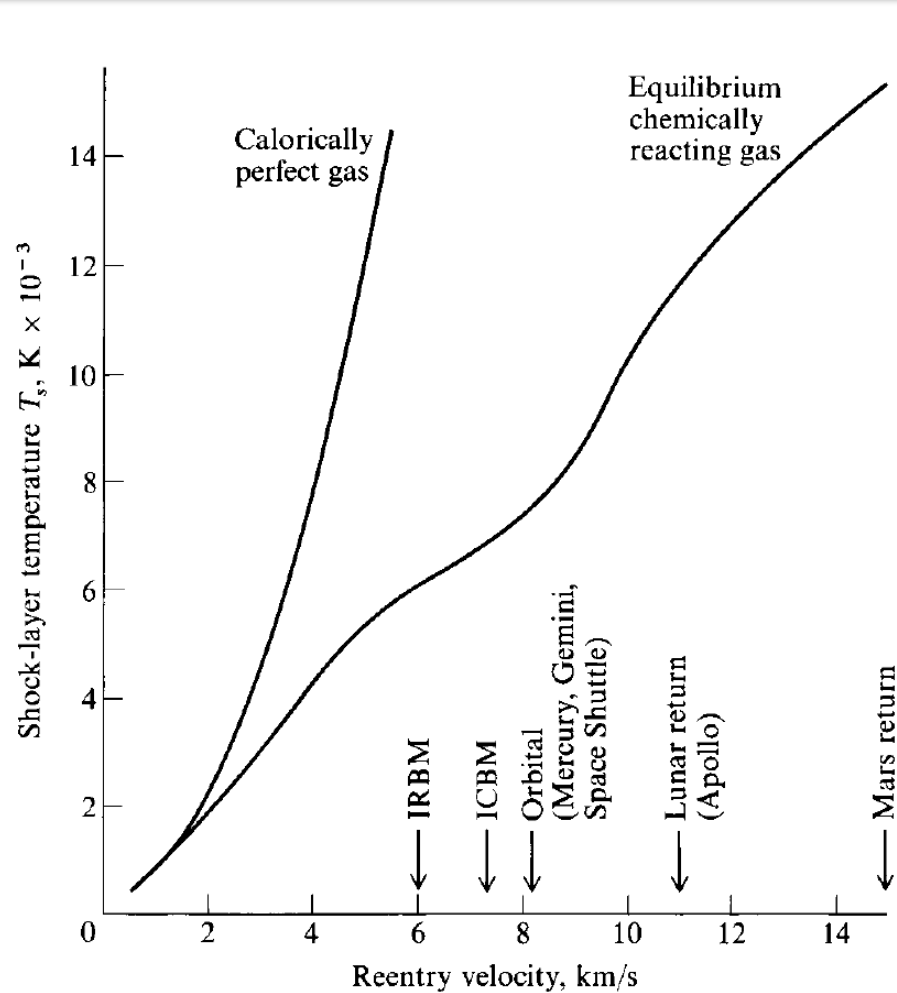
Shock layer temperature $T_s \approx 58300$ K

TOTALLY INCORRECT

Figures from:

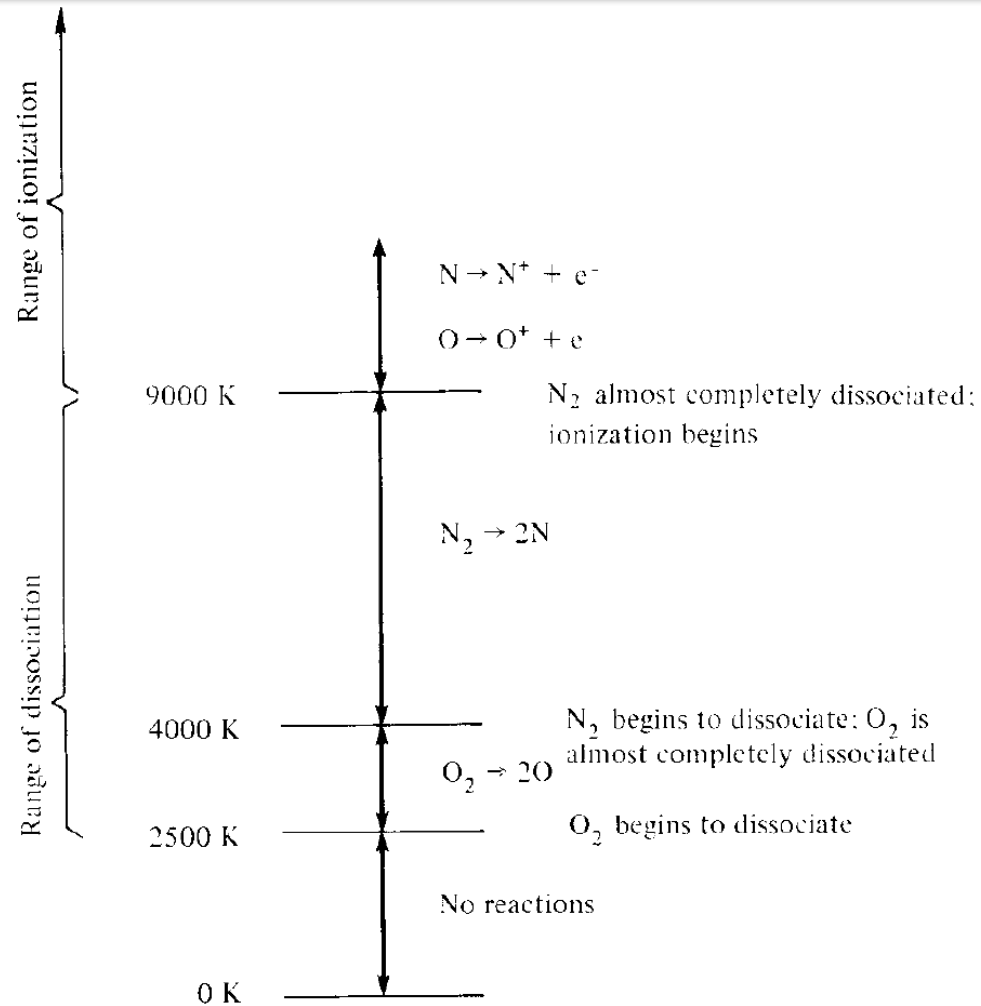
J.D Anderson, Hypersonic and High-Temperature Gas Dynamics Second Edition, AIAA, ISBN 978-1-56347-780-5

Calorically perfect gas vs local chemical equilibrium



52 km Altitude;

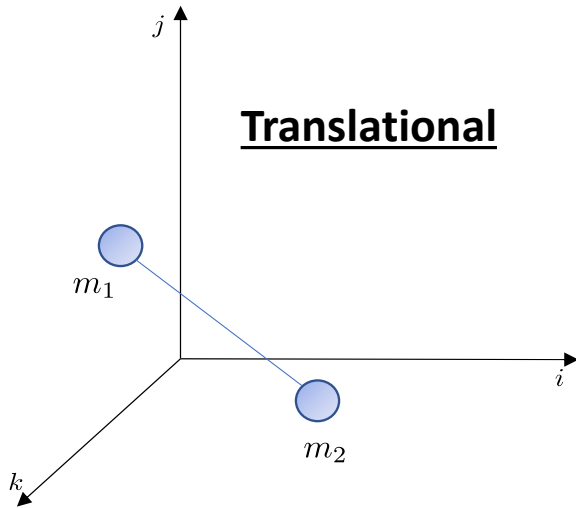
At very high temperature molecules dissociate and ionize



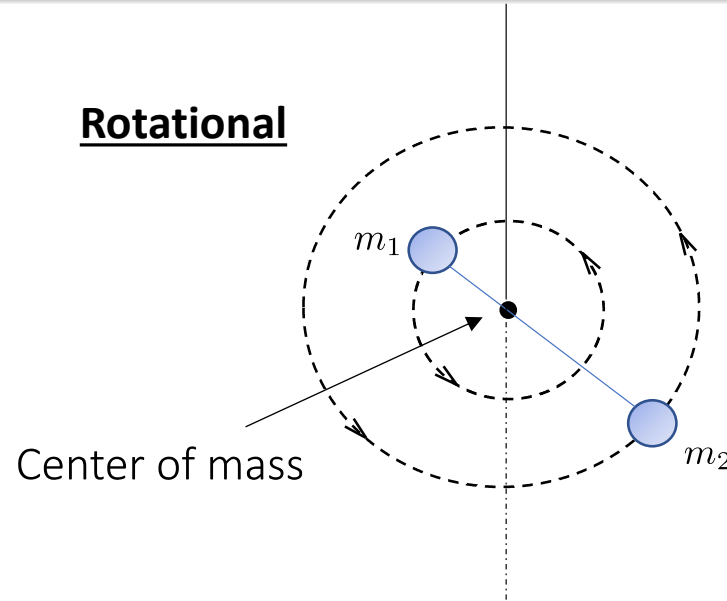
Local chemical equilibrium assumption $\longrightarrow T_s = 11600 \text{ K}$

Modes of Energy

Translational



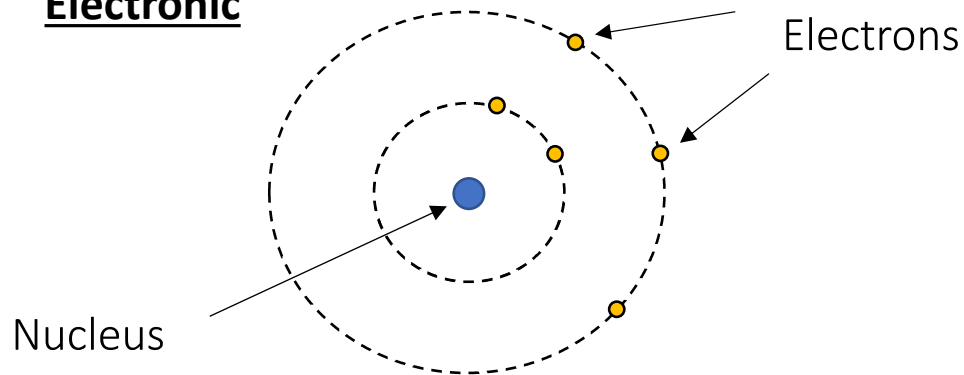
Rotational



Vibrational



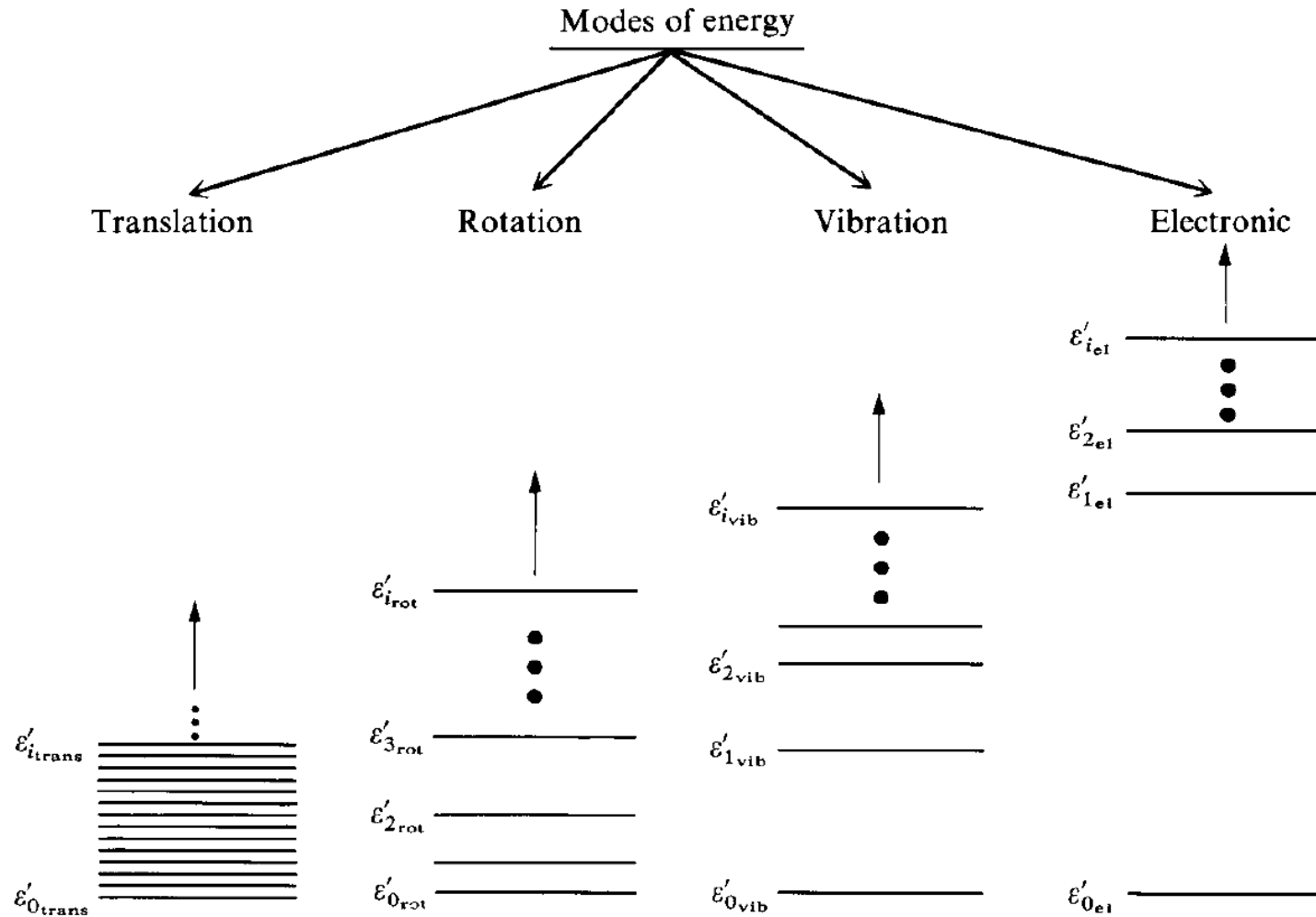
Electronic



A diatomic molecule stores energy as:

- **Translational: 3 degrees of freedom**
- **Rotational: 2 degrees of freedom**
- **Vibrational: 2 degrees of freedom**
- **Electronic: 2 degrees of freedom**

Modes of Energy



As a ladder, only discrete energy levels are allowed

- Translational
- Rotational
- Vibrational
- Electronic

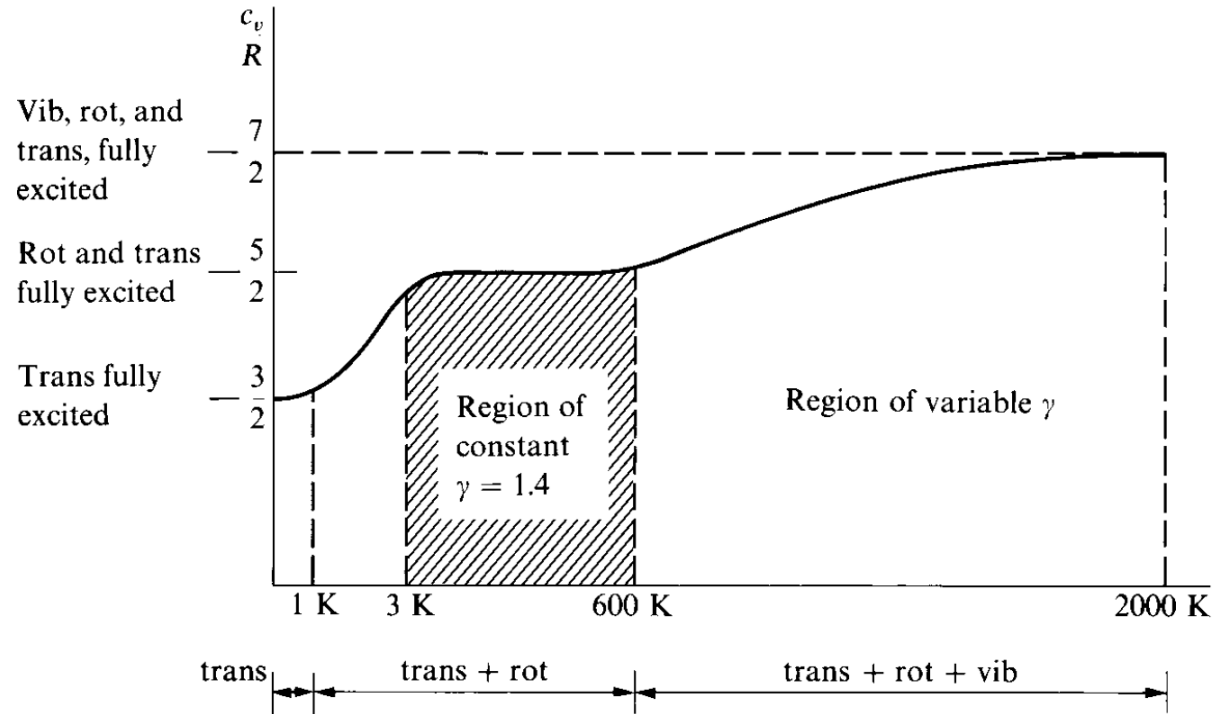
Increasing space
between levels

Figures from:

J.D Anderson, Modern Compressible Flow With
Historical Perspective, Fourth Edition McGraw-Hill
Education ISBN: 978-1-260-57082-3

Equilibrium thermally perfect gas

Theorem of equipartition of energy: each thermal degree of freedom contributes $RT/2$ to the energy



$T < 600 \text{ K} \quad e = e_{tr} + e_{rot}$
→
Classical equipartition theorem
 $e = \frac{3}{2}RT + RT$

$T > 600 \text{ K} \quad e = e_{tr} + e_{rot} + e_{vib} + e_{el}$
→
Statistical mechanics
 $e = \frac{3}{2}RT + RT + \frac{h\nu/kT}{e^{h\nu/kT} - 1}RT + e_{el}$

Thermochemical non-equilibrium

Excite different energy levels requires time

Each process has a different characteristic time

$$\tau_{tr} < \tau_{rot} < \tau_{vib} < \tau_{diss} < \tau_{el} < \tau_{ion}$$

τ_{flow} **Flow characteristic time**

If all $\tau \ll \tau_{flow}$ **The flow is in local equilibrium**

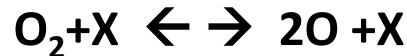
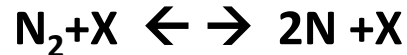
If all $\tau \gg \tau_{flow}$ **The flow is frozen**

If some $\tau \approx \tau_{flow}$ ***The flow is in
thermochemical non-equilibrium***

Multi-temperature 5 species Park model

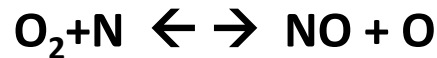
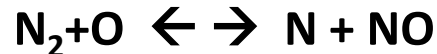
REACTIONS:

Dissociation



with $\text{X} = \text{N}_2, \text{O}_2, \text{NO}, \text{N}, \text{O}$

Zeldovich exchange reactions



$$\sum_{k=1}^K \nu'_{ki} \chi_k \rightleftharpoons \sum_{k=1}^K \nu''_{ki} \chi_k$$

Generic i^{th} reaction

$$\dot{\omega}_k = M_k \sum_{i=1}^I \nu_{ki} q_i$$

Chemical production rate of the k^{th} species

$$\nu_{ki} = \nu''_{ki} - \nu'_{ki}$$

Net stoichiometric coefficient

$$q_i = k_{fi} \prod_{k=1}^K [\chi_k]^{\nu'_{ki}} - k_{ri} \prod_{k=1}^K [\chi_k]^{\nu''_{ki}}$$

Rate of progress of the i^{th} reaction

Multi-temperature 5 species Park model

The two-temperature Park model assumes that the Arrhenius rate constants are functions of a geometrically averaged between the translational-rotational temperature (T) and the vibrational temperature (T_v) in the form:

$$T_a = T_v^q T^{1-q}$$

with q between 0.3 and 0.7

$$k_{fi} = A_i T^{\beta_i} \exp\left(\frac{-E_i}{R_c T_a}\right)$$

Arrhenius forward rate constant

$$k_{ri} = \frac{k_{fi}(T_a)}{K_{C_i}(T_a)}$$

reverse rate constant

$$\dot{\omega}_{LT,m} = \rho_m \frac{e_{vib,m}(T) - e_{vib,m}(T_{v,m})}{\tau_m}$$

Landau Teller evolution of the vibrational energy

τ_m Vibrational energy relaxation time (Millikan-White expression)

$$\dot{\omega}_{chem,m} = \frac{D_m}{2} \dot{\omega}_m$$

Preferential removal

Multi-temperature 5 species Park model

τ_m **Vibrational energy relaxation time (Millikan-White expression)**

$$\tau_{m,X}^{MW} = \frac{p_{atm}}{p} \exp [A_{m,X} (T^{-1/3} - B_{m,X}) - 18.42]$$

$$\tau_{m,X}^c = \frac{1}{n_m \sigma \sqrt{\frac{8\mathcal{R}T}{\pi \mathcal{M}_{m,X}}}}$$

Correction term

$$\tau_{m,X} = \tau_{m,X}^{MW} + \tau_{m,X}^c$$

$$\frac{1}{\tau_m} = \frac{1}{n_t} \sum_{s=1}^{N_s} \frac{n_s}{\tau_{m,X}}$$

MULTI-TEMPERATURE 5 SPECIES PARK MODEL¹

- 17 reactions + 3 transport equations for the vibrational energies
- Arrhenius type rate coefficients function of an effective temperature calculated as a geometrical mean of translational and vibrational temperatures
- Vibrational levels follow a Boltzmann distribution at temperature T_v
- Tuned on experimental measures
- Not computationally demanding
- It may fail when the conditions are far from those for which it was tuned

¹ C. Park, Nonequilibrium Hypersonic Aerothermodynamics, Wiley, New York, 1990

Thermochemical non-equilibrium models for air

MULTI-TEMPERATURE 5 SPECIES PARK MODEL¹

- 17 reactions + 3 transport equations for the vibrational energies
- Arrhenius type rate coefficients function of an effective temperature calculated as a geometrical mean of translational and vibrational temperatures
- Vibrational levels follow a Boltzmann distribution at temperature T_v
- Tuned on experimental measures
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5 SPECIES State-to-State (StS) MODEL²

- Detailed vibrational kinetics of molecules
- 68 and 47 vibrational levels for N_2 and O_2 respectively
- Thousands of elementary processes → High accuracy but huge computational cost

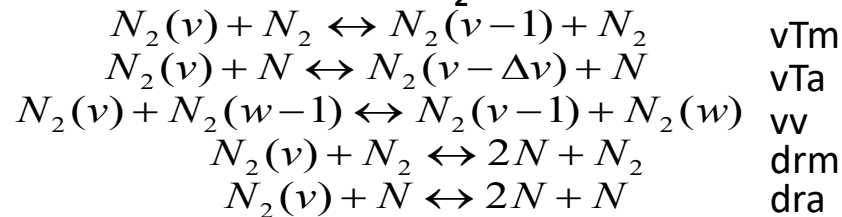
¹ C. Park, Nonequilibrium Hypersonic Aerothermodynamics, Wiley, New York, 1990

² M. Capitelli et al., Fundamentals Aspects of Plasma Chemical Physics: Kinetics, Springer Science & Business Media, 2015

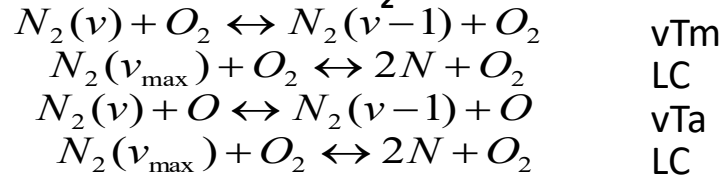
5 species State-to-State (StS) model

The State-to-State approach write a relaxation equation for each vibrational level so that it is possible to calculate the distribution of internal states when it departs from the Boltzmann one.

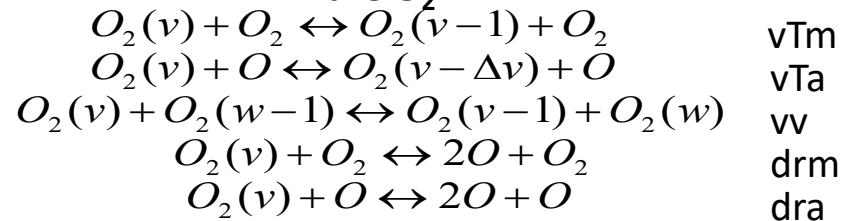
Pure N₂



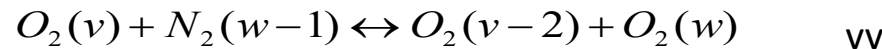
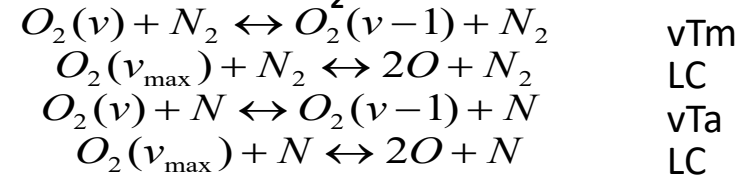
Mixed N₂



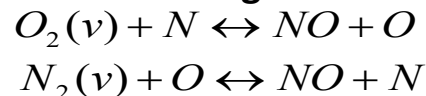
Pure O₂



Mixed O₂



Zeldovich exchange reactions



vTm/vTa: vibrational translational energy exchange with molecules/atoms;

vv: vibrational vibrational energy exchange

drm/dra: dissociation-recombination with molecules/atoms

LC: ladder climbing

Governing equations

Compressible Navier-Stokes (N-S) equations for a multicomponent mixture of reacting gases in thermochemical non-equilibrium for both Park and StS models

$$\frac{\partial \rho_{s,l}}{\partial t} + \nabla \cdot (\rho_{s,l} \mathbf{u} + \rho_{s,l} \mathbf{u}_{s,l}^D) = \dot{\omega}_{s,l} \quad \begin{array}{ll} s = 1, N_s & \text{number of species} \\ l = 1, V_s & \text{number of levels} \end{array}$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u} + p \bar{\bar{\mathbf{I}}} - \bar{\bar{\boldsymbol{\sigma}}}) = 0$$

$$\frac{\partial \rho E}{\partial t} + \nabla \cdot [(\rho E + p) \cdot \mathbf{u} - \mathbf{u} \cdot \bar{\bar{\boldsymbol{\sigma}}} + \mathbf{q}] = 0$$

$$\mathbf{q} = -\kappa_{tr} \nabla T - \sum_{m=1}^{N_m} \kappa_{vib,m} \nabla T_{vib,m} + \sum_{s=1}^{N_s} \sum_{l=1}^{V_s} \rho_{s,l} h_{s,l} \mathbf{u}_{s,l}^D$$

$$\frac{\partial \rho_m \varepsilon_{vib,m}}{\partial t} + \nabla \cdot [(\rho_m \varepsilon_{vib,m}) \cdot \mathbf{u} + \mathbf{q}_{vib,m}] = \dot{\omega}_{vib,m}$$

$$\mathbf{q}_{vib,m} = -\kappa_{vib,m} \nabla T_{vib,m} + \rho_m \varepsilon_{vib,m} \mathbf{u}_m^D$$

Transport properties of single species evaluated by using Gupta's curve fits [1].

Classical mixing rules are used for mixture properties.

[1] Gupta et al. A review of reaction rates and thermodynamic and transport properties for an 11-species air model for chemical and thermal nonequilibrium calculations to 30000 K, NASA. Reference. Publication. 1232. 1990

Governing equations

$$\rho_s \mathbf{u}_s^D = -\rho \mathcal{D}_s \nabla Y_s + \rho_s \mathbf{V}_c$$

Fick's law

$$\mathbf{V}_c = \sum_s \mathcal{D}_s \nabla Y_s$$

correction velocity

$$\mu_s = \left[\exp(C_{\mu_s}) \right] T^{(A_{\mu_s} \ln T + B_{\mu_s})}$$

Gupta's curve fits

$$\mu = \frac{\sum_{s=1}^{N_s} \chi_s \mu_s}{\sum_{r=1}^{N_s} \chi_r \phi_{sr}}$$

Wilke's mixing

$$\phi_{sr} = \frac{1}{\sqrt{8}} \left(1 + \frac{\mathcal{M}_s}{\mathcal{M}_r} \right)^{-1/2} \left[1 + \left(\frac{\mu_s}{\mu_r} \right)^{1/2} \left(\frac{\mathcal{M}_r}{\mathcal{M}_s} \right)^{1/4} \right]^2$$

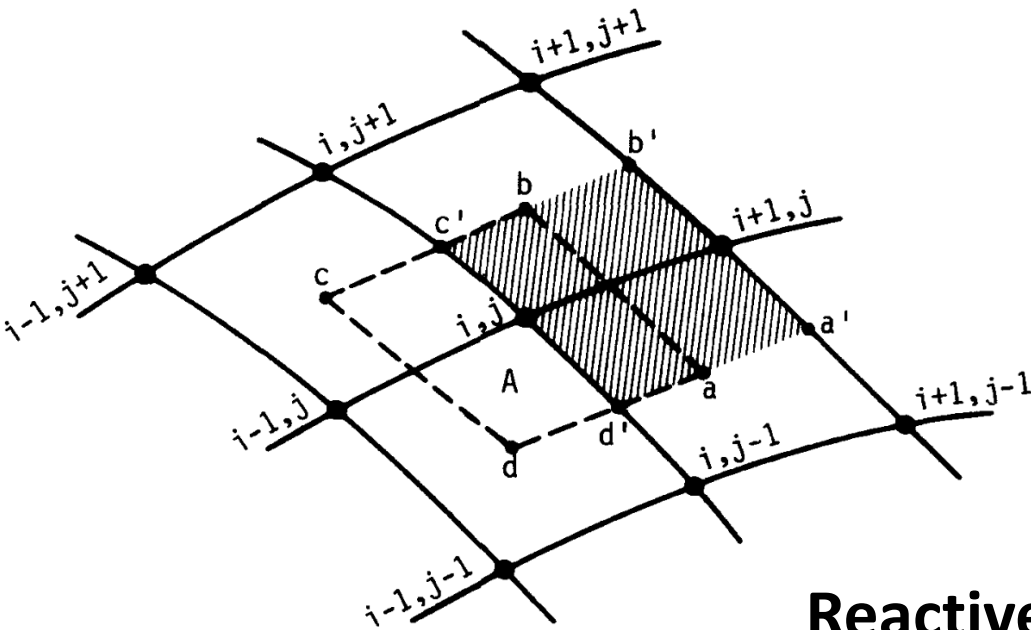
$$\lambda_s = \mu_s \left(\frac{5}{4} + \alpha_s \right) \frac{\mathcal{R}}{\mathcal{M}_s}$$
$$\lambda_{m,vib} = \mu_m c_{v,vib}$$

Eucken's formula

Numerical method

$$V_{i,j} \frac{d\mathbf{U}_{i,j}}{dt} + \sum_{\text{Faces}} \mathbf{F}_{num} \cdot \mathbf{n} \Delta S = V_{i,j} \mathbf{W}_{i,j}$$

***Cell-centered Finite Volume
Space discretization on a
Multi-block structured mesh***



$$\mathbf{F}_{num} = \mathbf{F}_{E,num} - \mathbf{F}_{V,num}$$

Reactive Navier-Stokes equations:

- Advection and pressure term (hyperbolic)
- Shear-stress, heat flux terms (diffusive)
- Chemical source terms (stiffness)

$$V_{i,j} \frac{d\mathbf{U}_{i,j}}{dt} + \sum_{Faces} \mathbf{F}_{num} \cdot \mathbf{n} \Delta S = V_{i,j} \mathbf{W}_{i,j}$$

$$\mathbf{F}_{num} = \mathbf{F}_{E,num} - \mathbf{F}_{V,num}$$

Solution strategy:

- Operator splitting approach: Frozen step + Chemical step
 - ✓ Frozen step: Method of Lines:
 - Space discretization + Time integration
 - Space discretization: Inviscid & Viscous terms scheme
 - Time integration: Runge-Kutta scheme
 - ✓ Chemical step: implicit scheme for stiff terms

Frozen step: inviscid flux space discretization

$$V_{i,j} \frac{d\mathbf{U}_{i,j}}{dt} + \sum_{Faces} \mathbf{F}_{num} \cdot \mathbf{n} \Delta S = 0 \quad \text{Frozen equation}$$

Semi-Discrete Schemes or Method of Lines

$$\frac{d\mathbf{U}_{i,j}}{dt} = -\frac{1}{V_{i,j}} \sum_{Faces} (\mathbf{F}_{E,num} - \mathbf{F}_{V,num}) \cdot \mathbf{n} \Delta S \quad \text{ODE solved with an explicit Runge-Kutta schemes}$$

$\mathbf{F}_{E,num}$

Methods for solving non-linear hyperbolic conservation laws:

Steger and Warming Flux Vector Splitting: Upwind discretization

The discretisation of the equations on a mesh is performed according to the direction of propagation of information on that mesh.

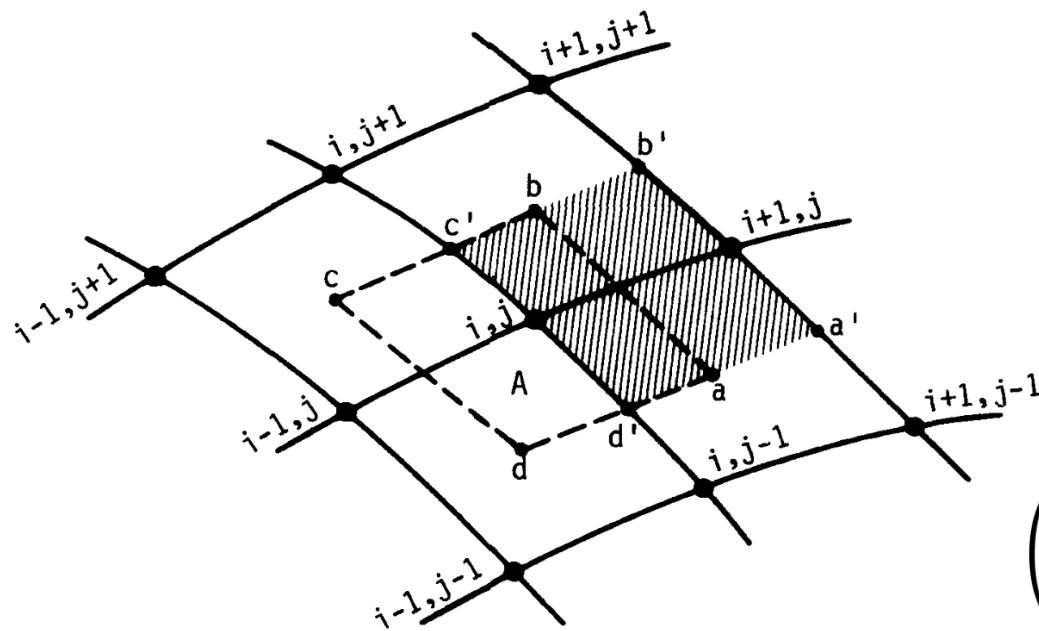
MUSCL reconstruction

Frozen step: viscous schemes

Viscous terms involve gradients that have to be determined on the cell faces

Due to their dissipative nature central differences are used

A good procedure for generalized curvilinear coordinates is to apply the Gauss divergence theorem



$$\int_{V_0} \nabla u dV = \oint_{S_0} u d\mathbf{S}$$

$$\nabla u = \frac{1}{\Delta V} \sum_{i=faces} u_i \Delta \mathbf{S}_i$$

$$\begin{pmatrix} u_x \\ u_y \end{pmatrix} = \frac{1}{\Delta V} \begin{pmatrix} \sum_{i=faces} u_i \Delta S_{x,i} \\ \sum_{i=faces} u_i \Delta S_{y,i} \end{pmatrix}$$

Operator splitting approach

$$\frac{\partial}{\partial t} \int_{V_0} \mathbf{U} dV + \oint_{S_0} \mathbf{F} \cdot \mathbf{n} dS = 0 \quad \text{Frozen step}$$

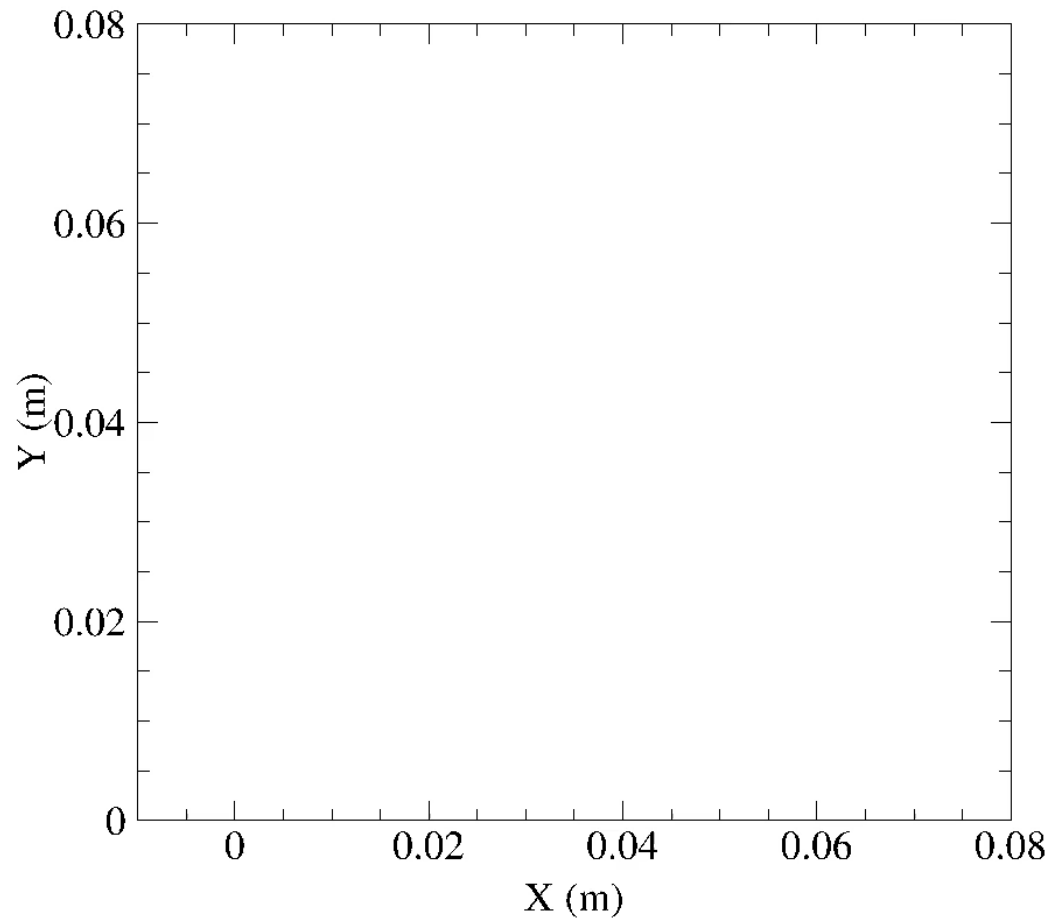
Inviscid flux: Flux Vector Splitting of Steger and Warming or AUSM with MUSCL approach for higher order accuracy;
Runge-Kutta scheme up to third order for time integration

Viscous flux: gradients of the primitive variables are evaluated by applying Gauss theorem

$$\frac{\partial}{\partial t} \int_{V_0} \mathbf{U} dV = \int_{V_0} \mathbf{W} dV \quad \text{Chemical step}$$

$$\Delta t_c^{(v)} = \Delta t_f / n \quad \text{Sub-time step}$$

Operator splitting approach



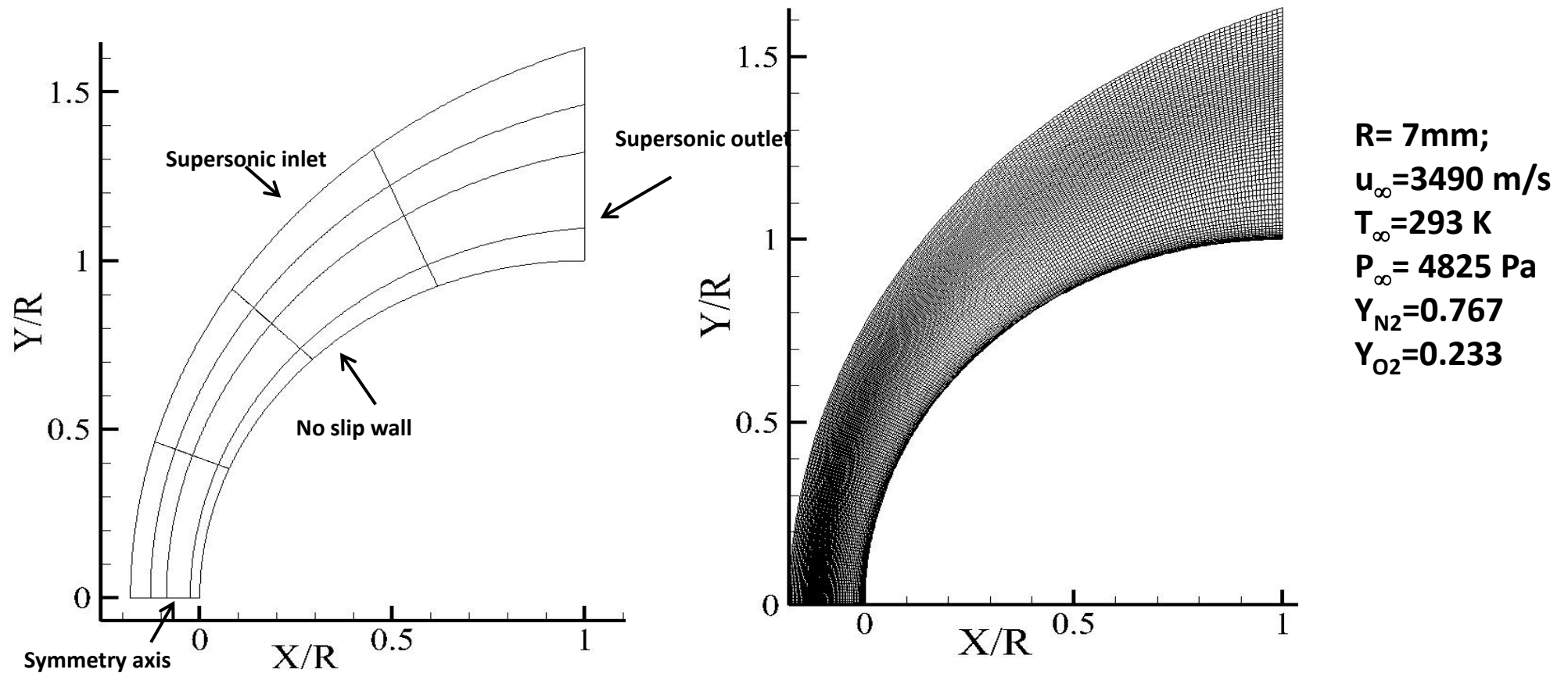
$$\frac{\partial \mathbf{y}}{\partial t} = \mathbf{P} - \mathbf{L}\mathbf{y} \quad \mathbf{y} = \{\rho_i\}_{0 \leq i \leq N}$$

$\mathbf{P}$ is a vector and $\mathbf{L}$ a diagonal matrix.
 P_i and $L_i y_i$ (non-negative) represent, respectively, production and loss terms for component y_i

$$y_i^k(t + \Delta t_c^{(v)}) = \frac{\Delta t_c^{(v)} P_i(\mathbf{y}^{k-1}) + y_i(t)}{1 + \Delta t_c^{(v)} L(\mathbf{y}^{k-1})}$$

Gauss-Seidel iterative scheme

Flow past a sphere: Nonaka* test case

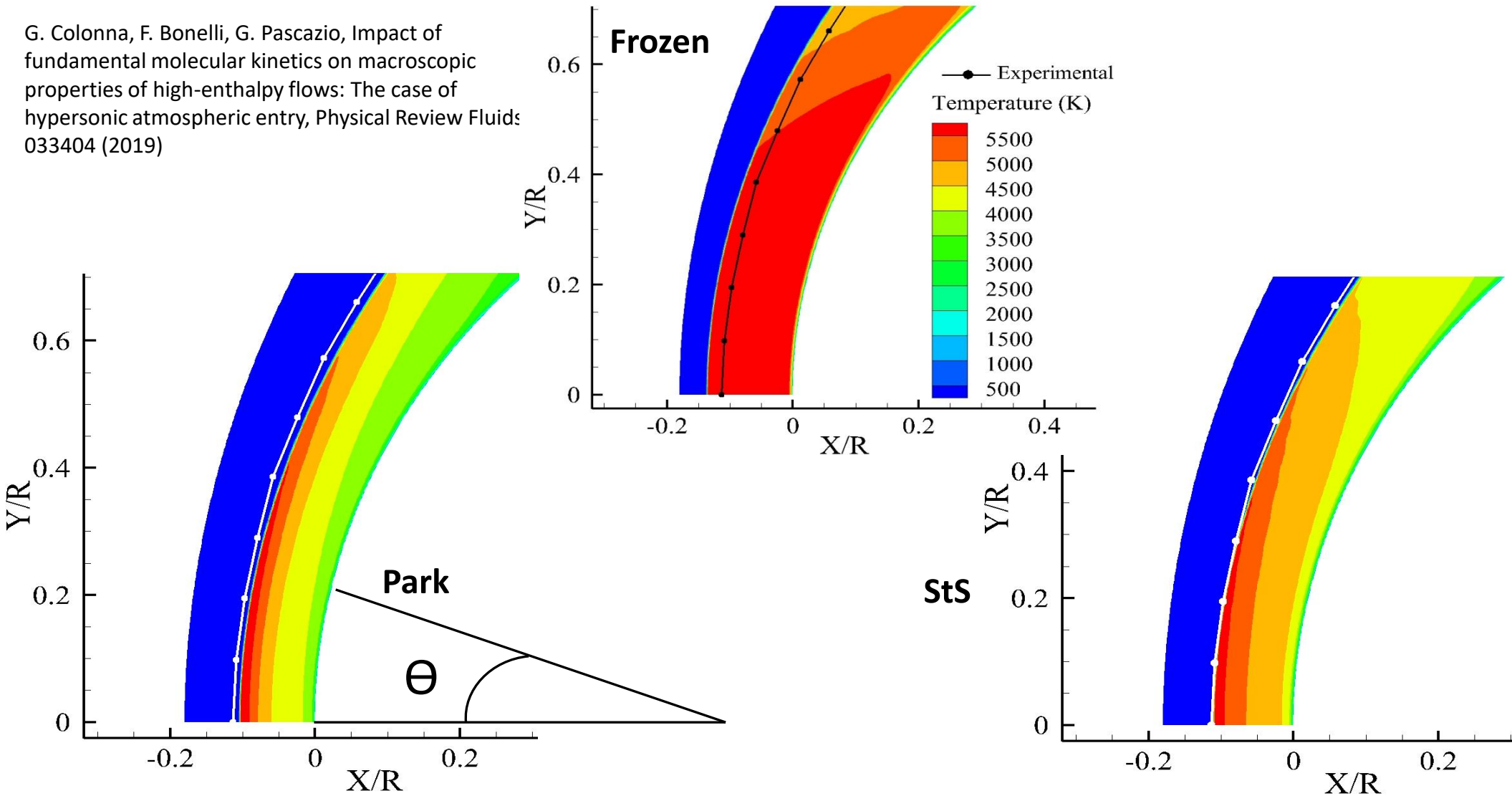


Computational domain, with an example of 4 x4 MPI partitioning, along with boundary conditions (left). 152x392 computational grid shown every 2 grid points (right).

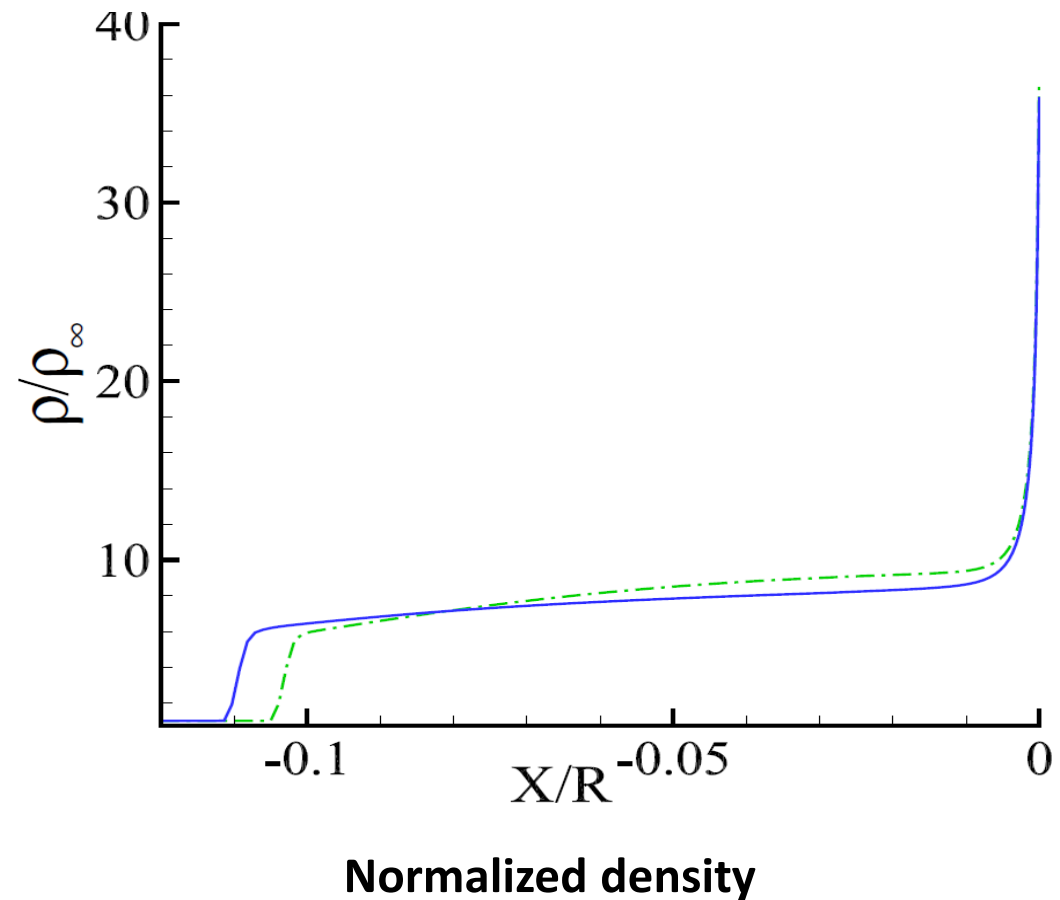
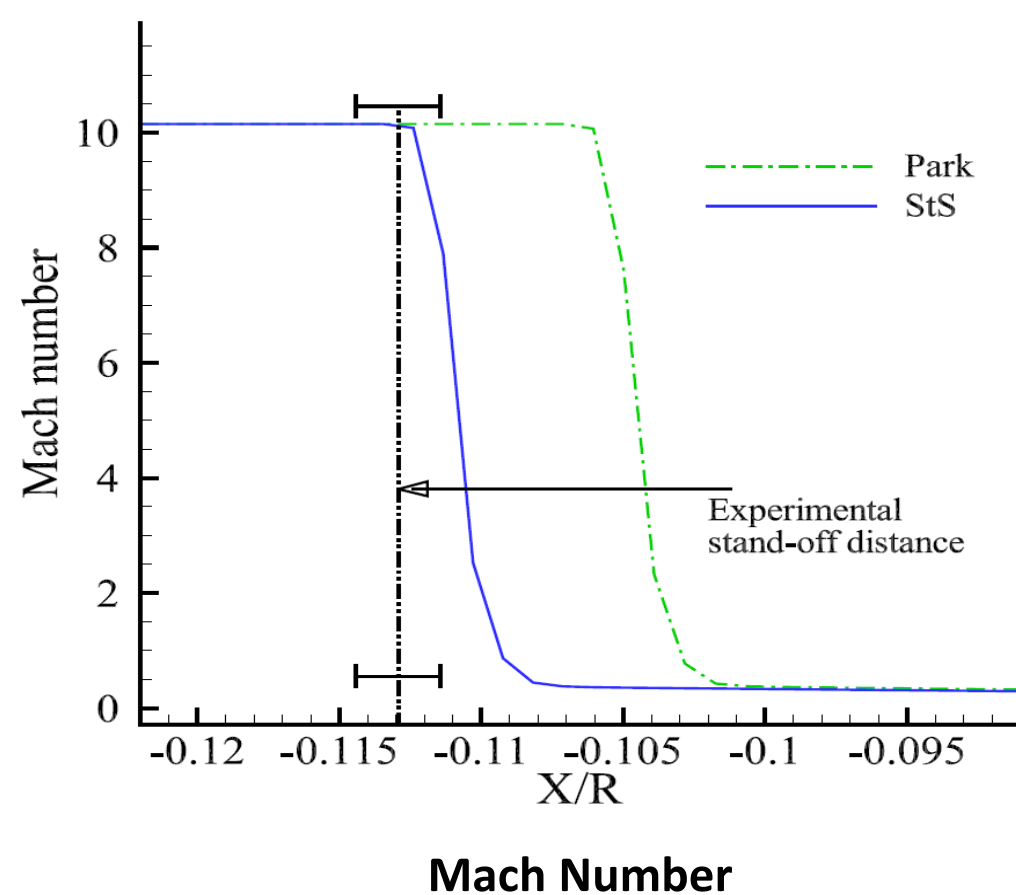
*S. Nonaka et al. ,JTHT 14 (2), 2000

Nonaka test case

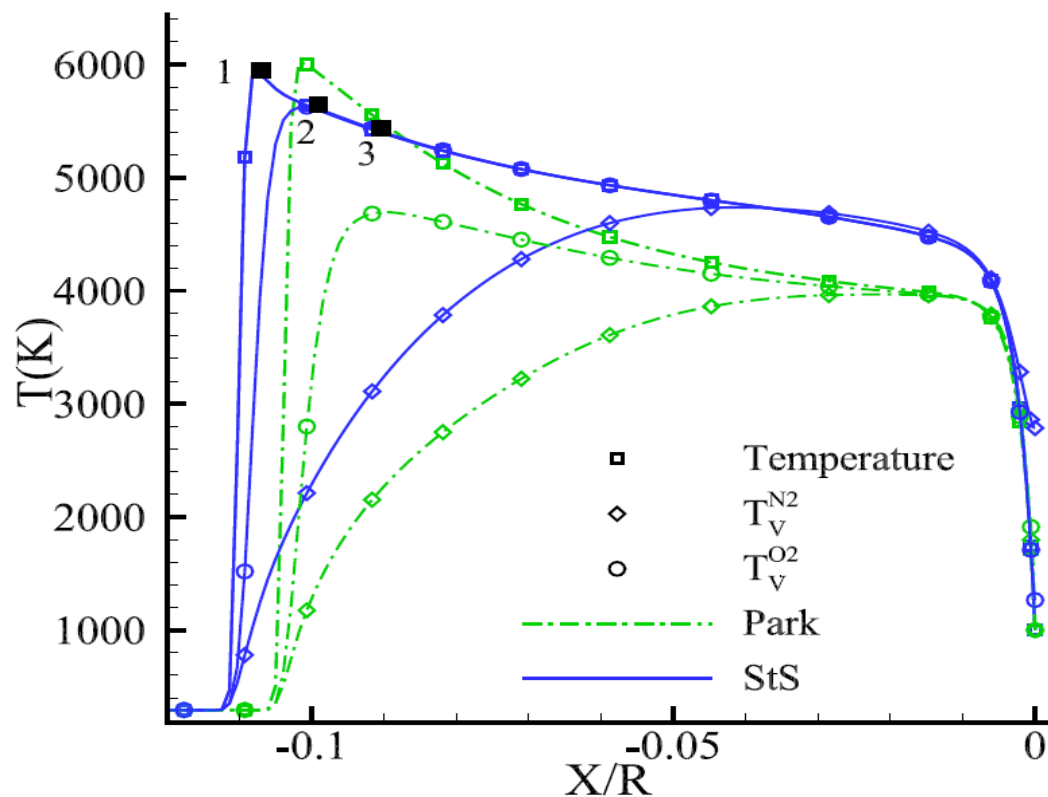
G. Colonna, F. Bonelli, G. Pascazio, Impact of fundamental molecular kinetics on macroscopic properties of high-enthalpy flows: The case of hypersonic atmospheric entry, Physical Review Fluids: 033404 (2019)



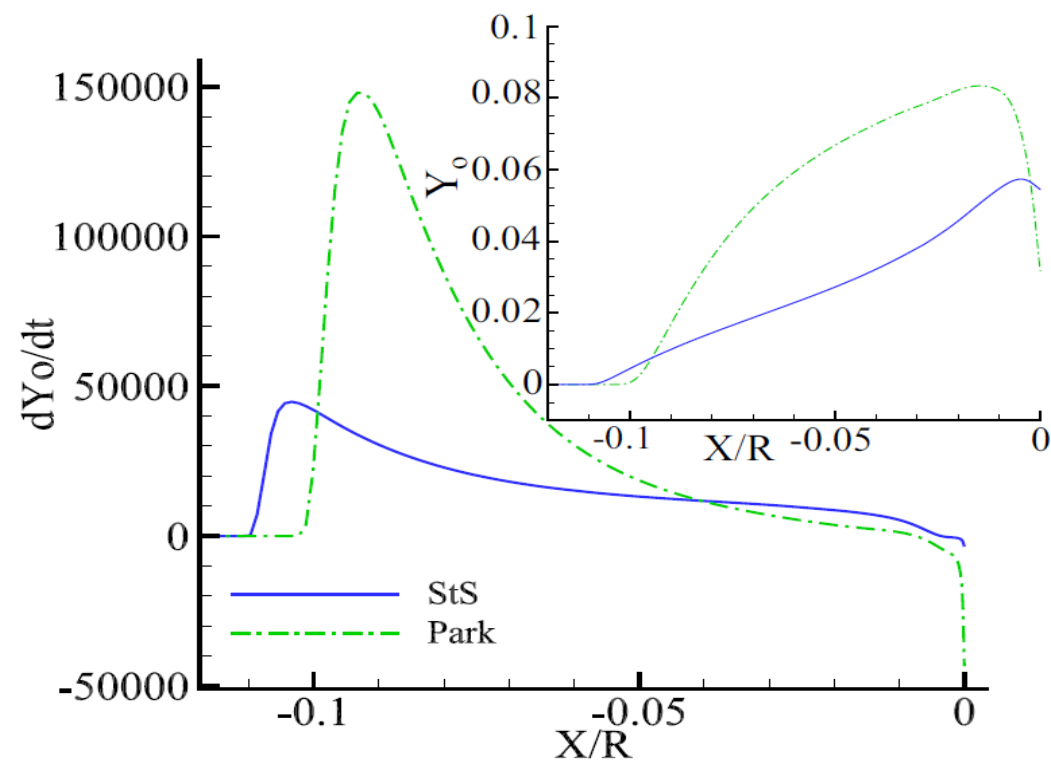
Nonaka test case: stagnation line profiles



Nonaka test case: stagnation line profiles

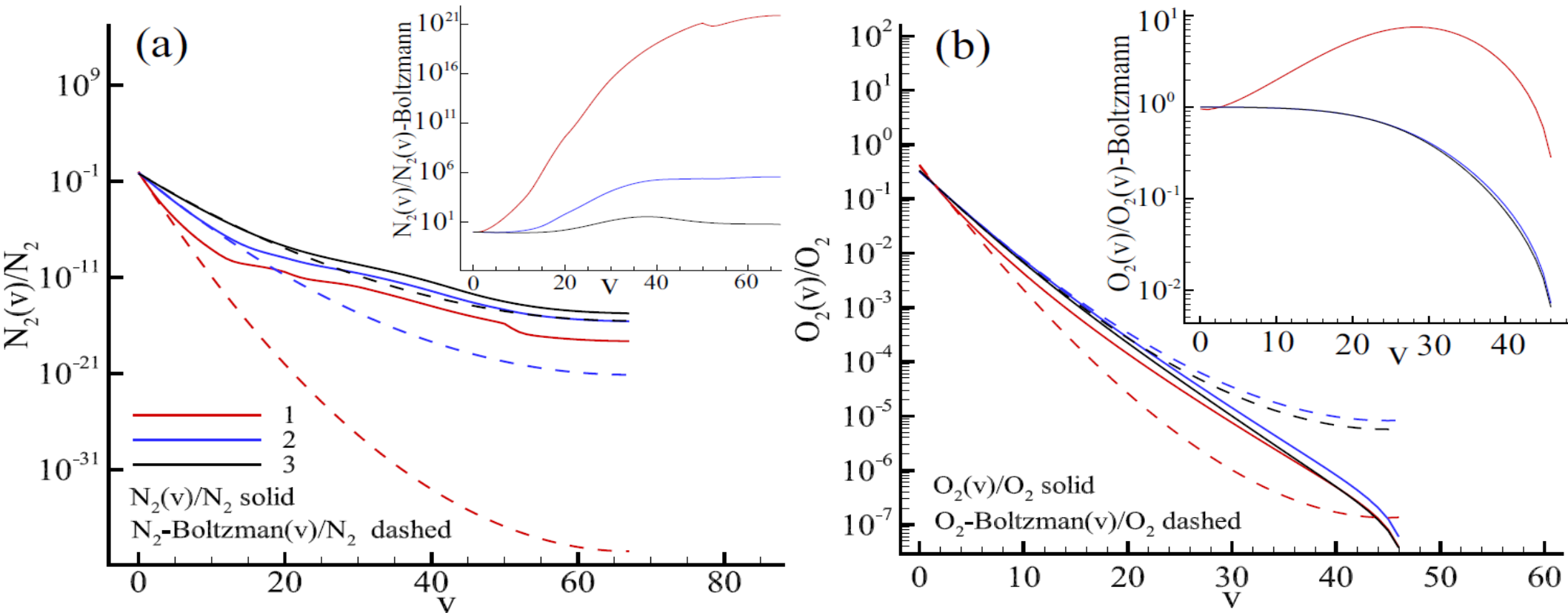


Temperatures



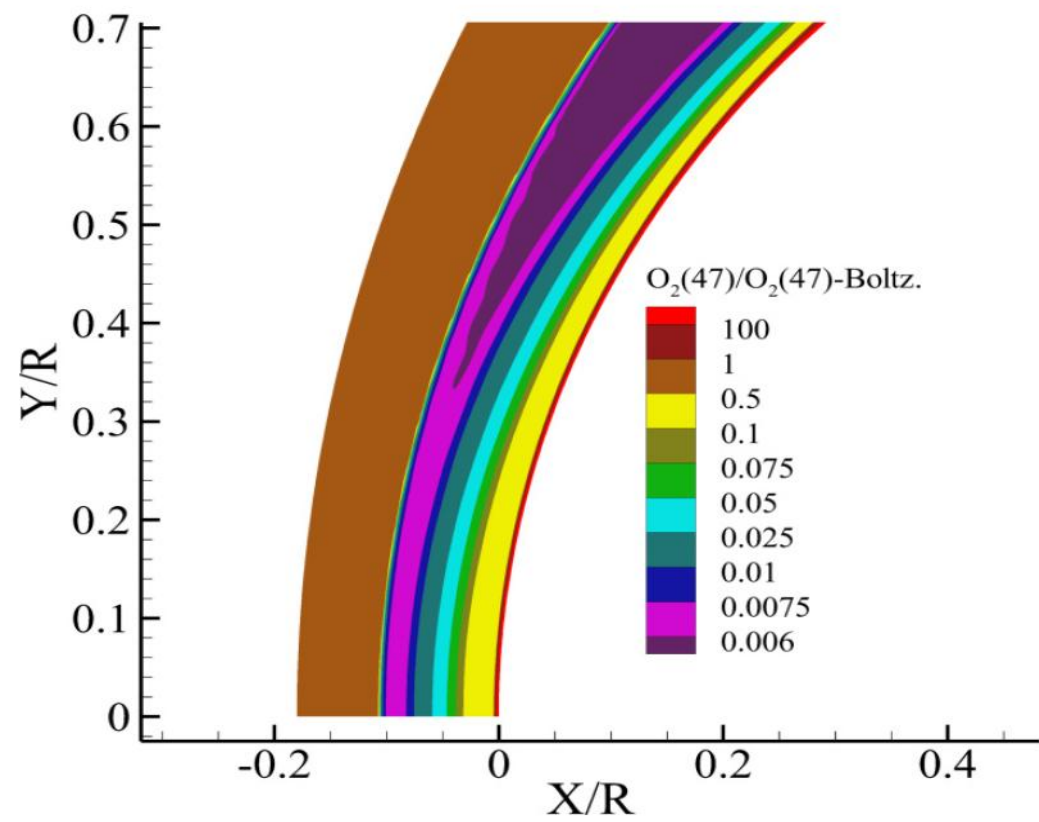
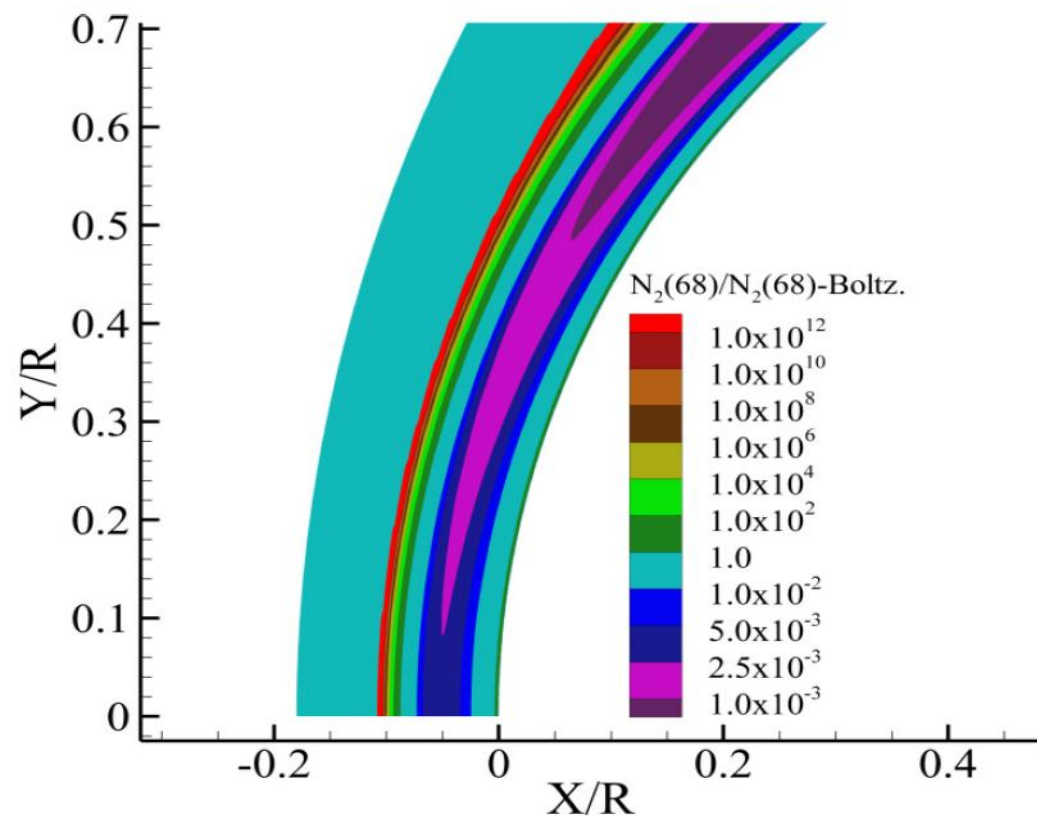
Oxygen atom reaction rate

Nonaka test case: vibrational distributions along the stagnation line



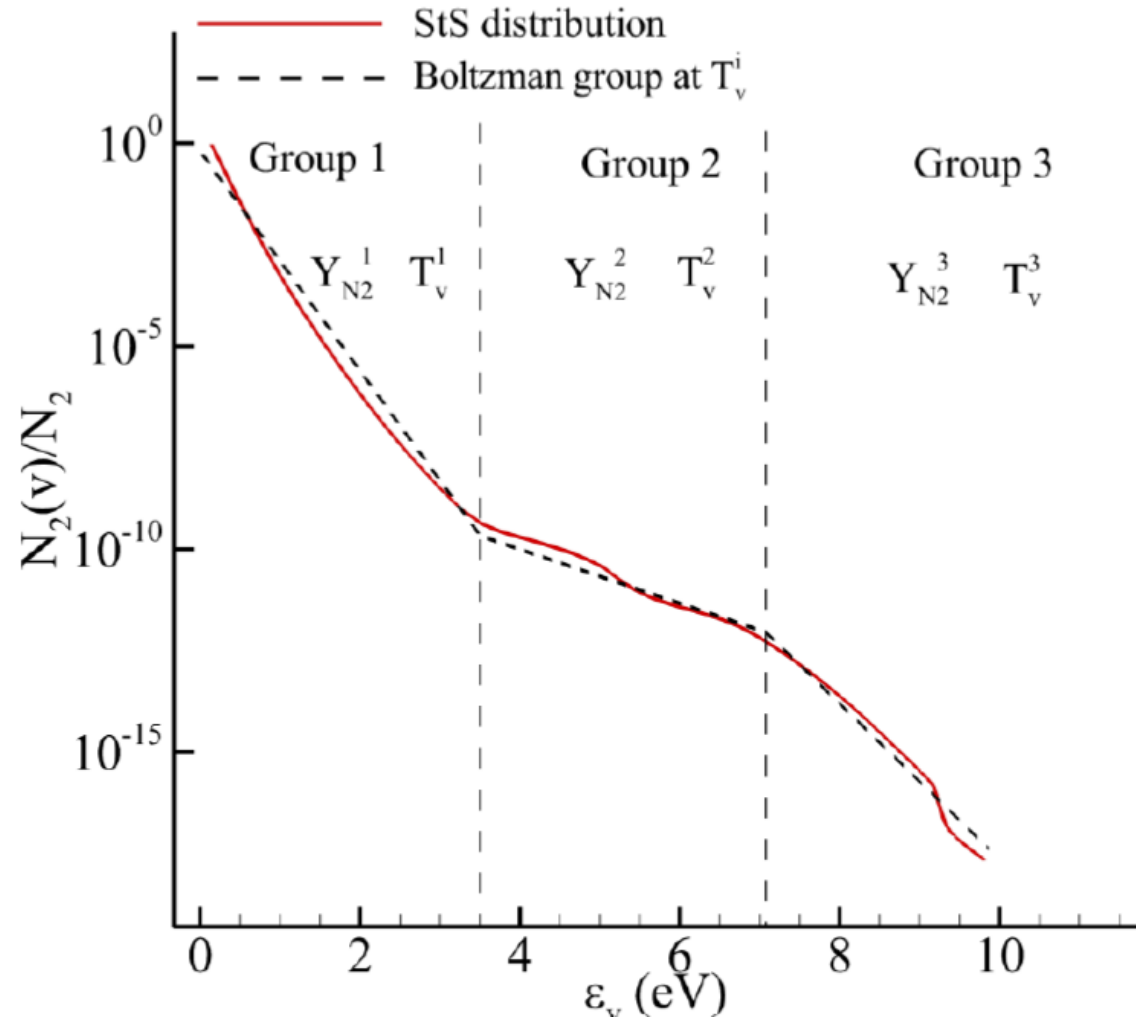
Just downstream of the bow shock distributions depart from the Boltzmann one

Nonaka test case: highest vibrational level contour plot



Actual to Boltzmann ratio for the highest vibrational level

Multi-internal-temperature kinetic model (MiT)

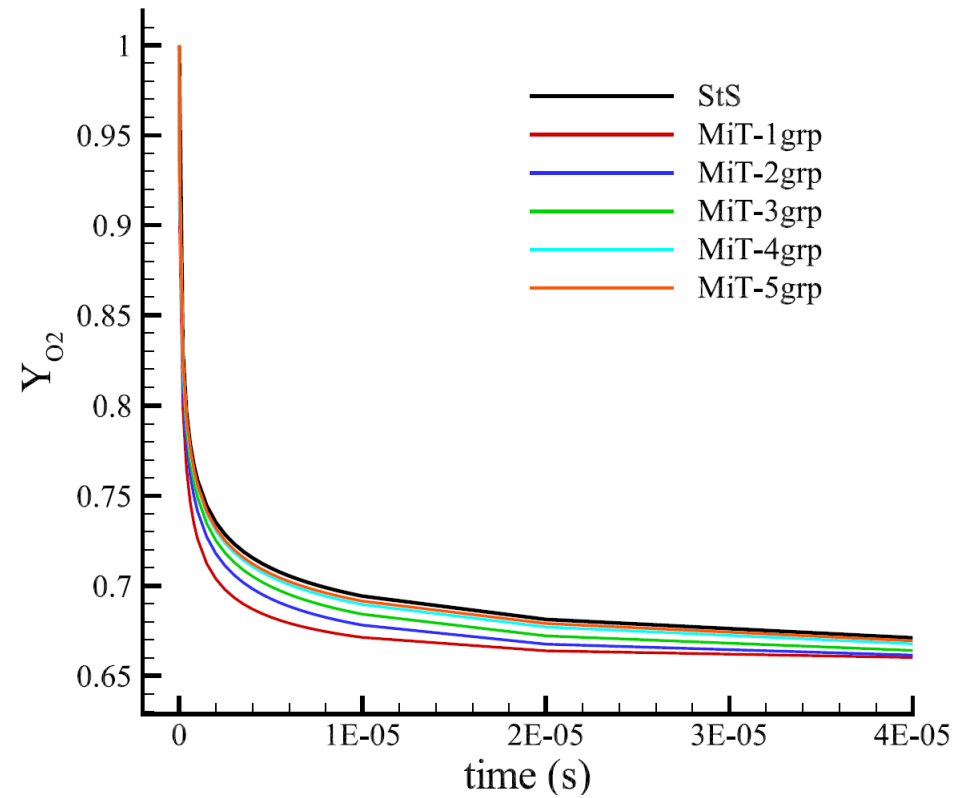
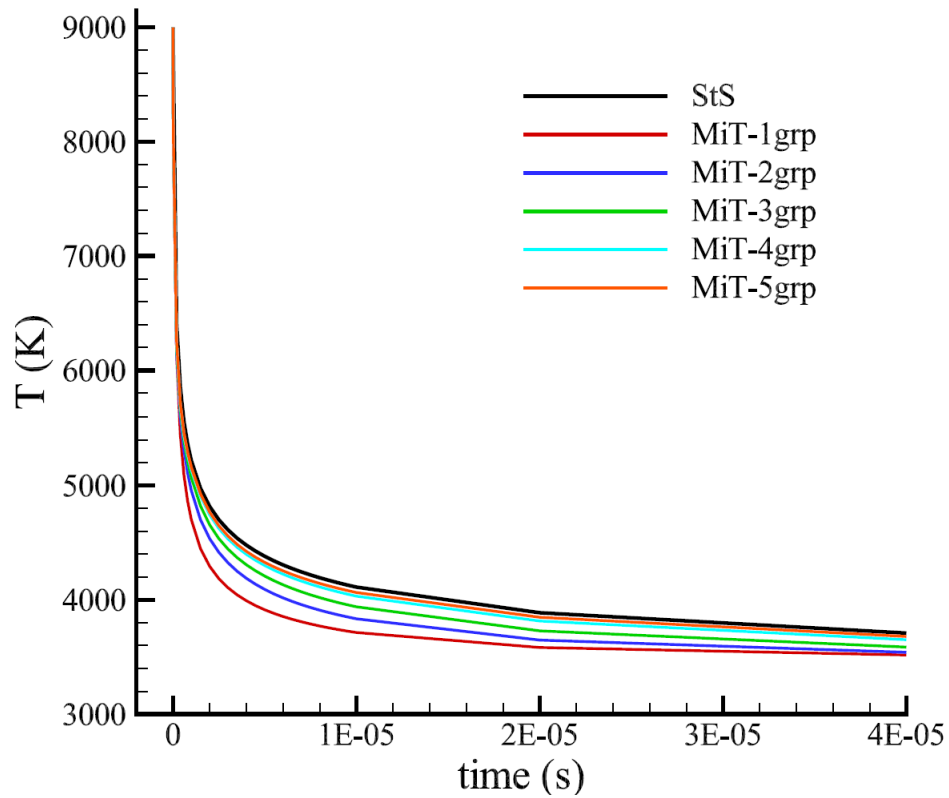


Guy A, Bourdon A, Perrin M-Y. Consistent multi-internal-temperatures models for nonequilibrium nozzle flows. Chem Phys 2013;420:15–24.

Bonelli, F., Ninni, D., Narracci, A., Colonna, G., & Pascazio, G. (2025). Assessment of a consistent multi-internal-temperature kinetic model for hypersonic neutral air flows using a finite volume solver. Computers & Fluids, 106796.

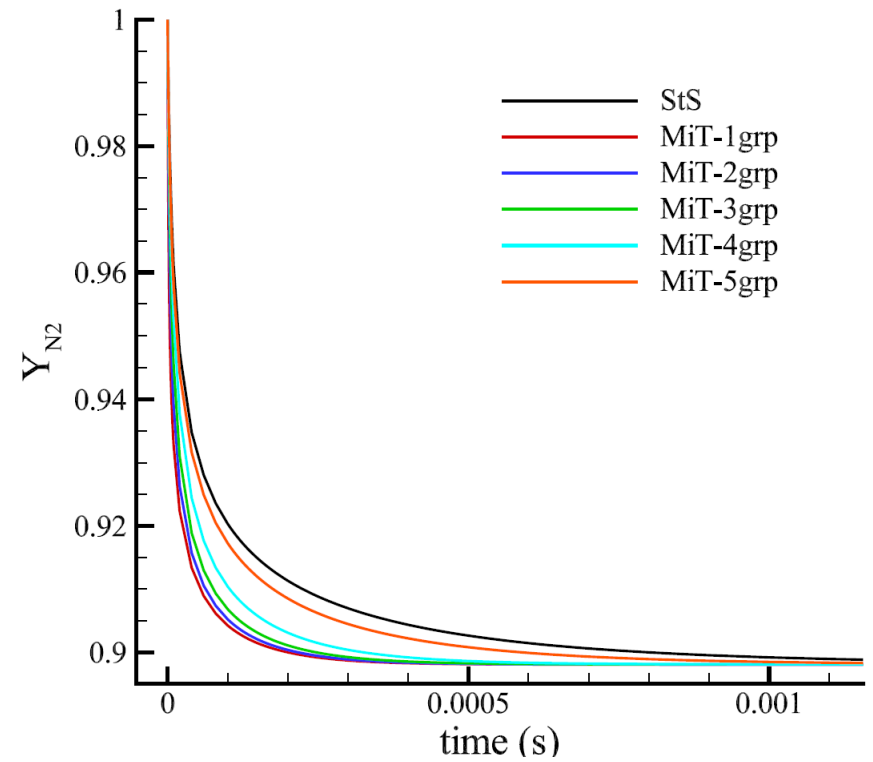
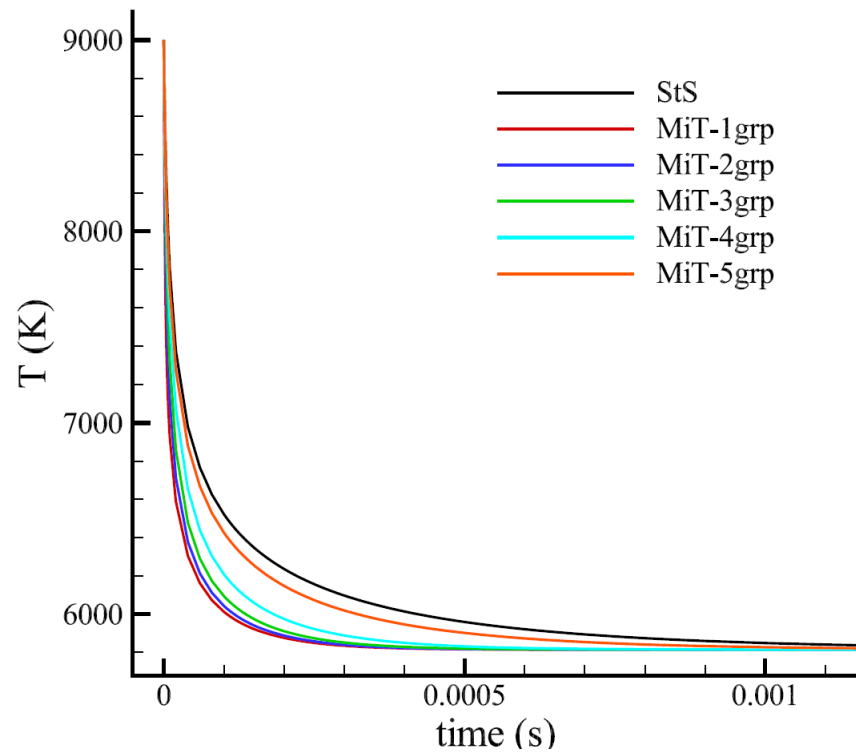
MiT: oxygen 0D reactor

$$Y_{O_2} = 1; T = 9000 \text{ K}; p = 10^5 \text{ Pa}$$

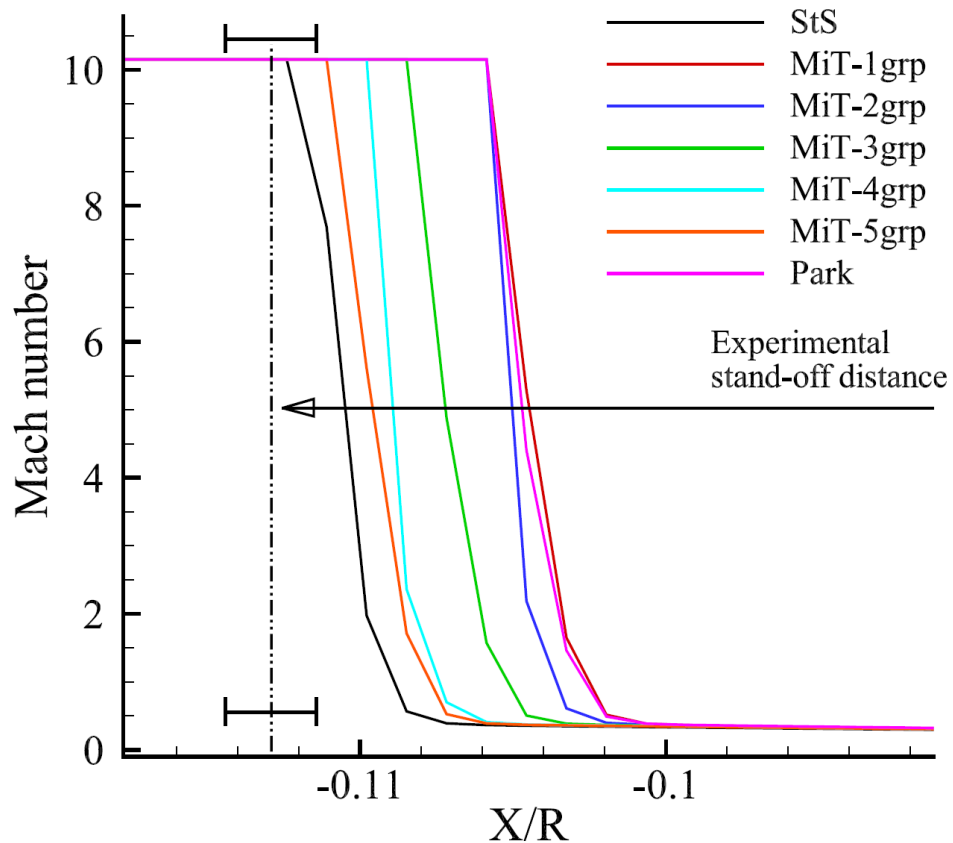


MiT: nitrogen 0D reactor

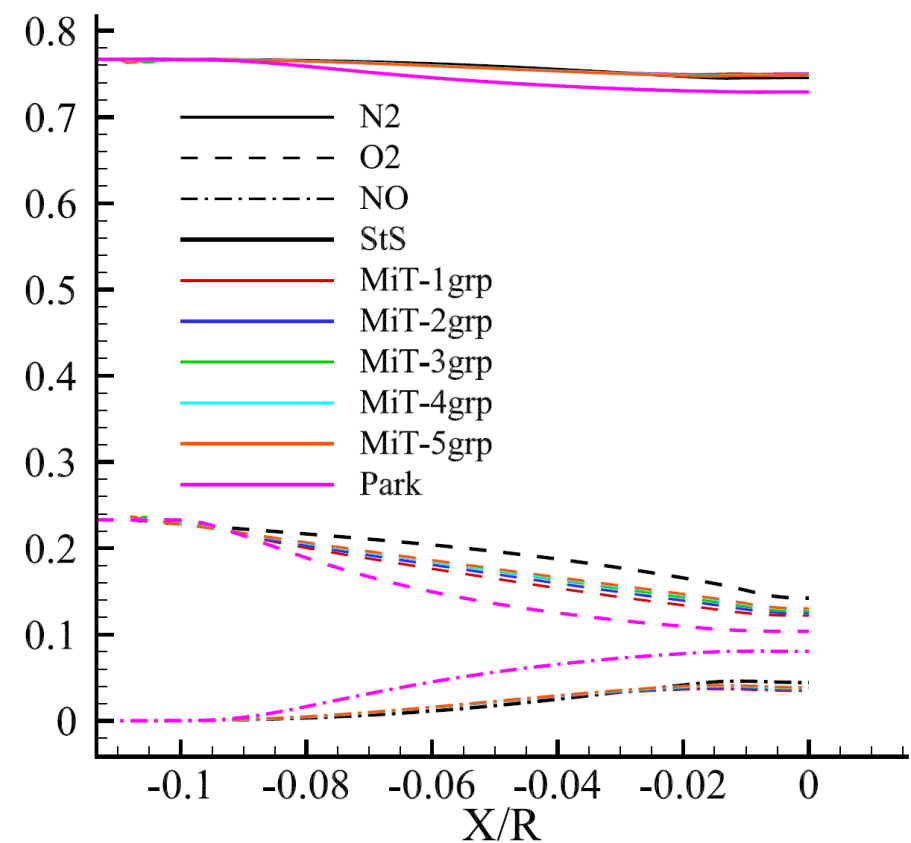
$$Y_{N_2} = 1; T = 9000 \text{ K}; p = 10^5 \text{ Pa}$$



MiT grouping strategy 2: Nonaka test case



Mach Number



Mass fractions

MiT: computational performance

0D CPU speed-up, grouping strategy 2, with $N_{kin} = 1$ and 8 Gauss–Seidel inner iterations. One core AMD Ryzen Threadripper 1920X processor.

	StS	MiT-1grp	MiT-2grp	MiT-3grp	MiT-4grp	MiT-5grp
TpI (s)	$1.70 \cdot 10^{-2}$	$1.03 \cdot 10^{-5}$	$1.39 \cdot 10^{-5}$	$1.67 \cdot 10^{-5}$	$2.10 \cdot 10^{-5}$	$2.46 \cdot 10^{-5}$
speed-up		$1.65 \cdot 10^3$	$1.22 \cdot 10^3$	$1.02 \cdot 10^3$	$8.07 \cdot 10^2$	$6.90 \cdot 10^2$

2D CPU speed-up, grouping strategy 2, on a 57×98 computational cells grid with $N_{kin} = 4$ and 8 Gauss–Seidel inner iterations. One core AMD Ryzen Threadripper 1920X processor.

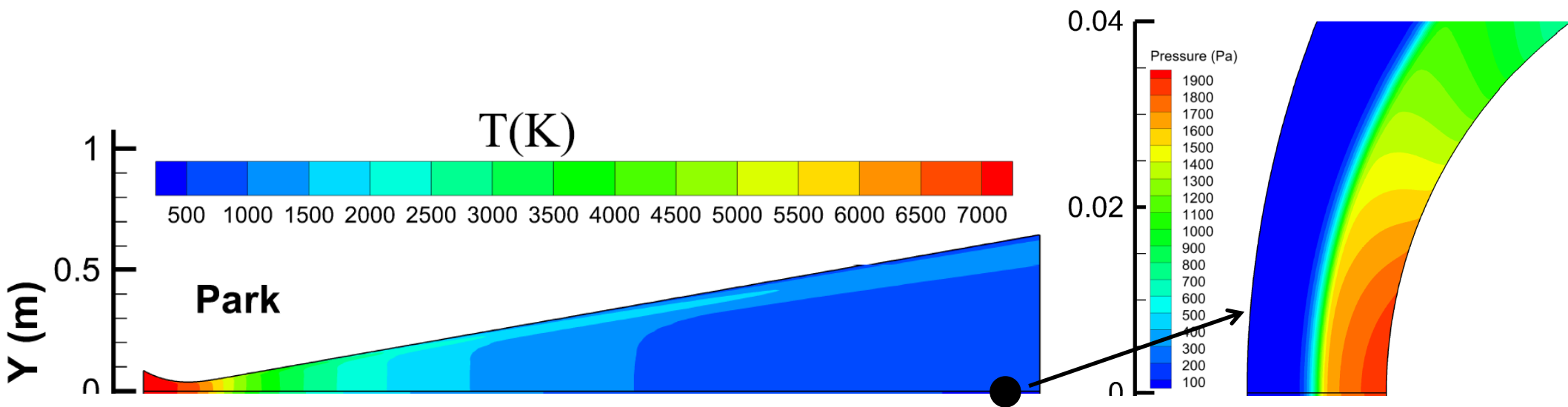
	StS	MiT-1grp	MiT-2grp	MiT-3grp	MiT-4grp	MiT-5grp
TpI (s)	$3.90 \cdot 10^2$	$7.01 \cdot 10^{-2}$	$1.13 \cdot 10^{-1}$	$1.74 \cdot 10^{-1}$	$2.65 \cdot 10^{-1}$	$3.96 \cdot 10^{-1}$
speed-up		$5.56 \cdot 10^3$	$3.45 \cdot 10^3$	$2.24 \cdot 10^3$	$1.47 \cdot 10^3$	$9.84 \cdot 10^2$

2D GPU speed-up, grouping strategy 2, on a 228×392 computational cells grid with $N_{kin} = 1$ and 8 Gauss–Seidel inner iterations. One NVIDIA A100 GPU on CINECA Leonardo Booster partition.

	StS	MiT-1grp	MiT-2grp	MiT-3grp	MiT-4grp	MiT-5grp
TpI (s)	$9.28 \cdot 10^{-1}$	$4.20 \cdot 10^{-3}$	$1.29 \cdot 10^{-2}$	$1.65 \cdot 10^{-2}$	$2.20 \cdot 10^{-2}$	$2.38 \cdot 10^{-2}$
speed-up		$2.21 \cdot 10^2$	$7.19 \cdot 10^1$	$5.62 \cdot 10^1$	$4.21 \cdot 10^1$	$3.90 \cdot 10^1$

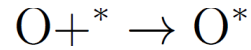
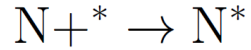
CIRA: SCIROCCO Plasma Wind Tunnel test case

$H_0 = 23.8 \text{ MJ/kg}$ $P_0 = 2.5 \text{ bar}$

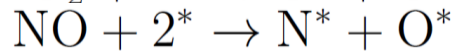
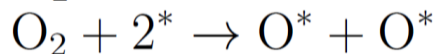
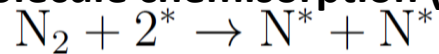


Finite rate catalysis model

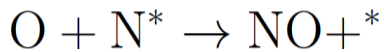
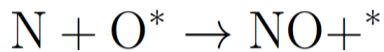
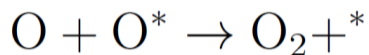
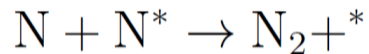
atom chemisorption (ch)



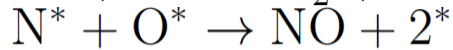
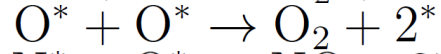
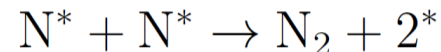
molecule chemisorption (chdm)



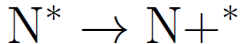
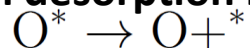
Eley-Rideal mechanisms (ER)



Langmuir-Hinshelwood mechanisms (LH)



thermal desorption mechanism (td)



$$\gamma_{AB} = \frac{\text{Flux of atoms recombining at the surface}}{\text{Flux of atoms impinging on the surface}}$$

$$\gamma_{NN} = \frac{2(-[N_2][S]^2 k_{chdm}^{N_2} + [N][N^*] k_{ER}^{NN} + [N^*]^2 k_{LH}^{NN})}{Z_N}$$

$$\gamma_{OO} = \frac{2(-[O_2][S]^2 k_{chdm}^{O_2} + [O][O^*] k_{ER}^{OO} + [O^*]^2 k_{LH}^{OO})}{Z_O}$$

$$\gamma_{NO} = \frac{(-[NO][S]^2 k_{chdm}^{NO} + [N][O^*] k_{ER}^{NO} + [O][N^*] k_{ER}^{ON} + [N^*][O^*] k_{LH}^{NO})}{Z_N}$$

$$\gamma_{ON} = \frac{(-[NO][S]^2 k_{chdm}^{NO} + [N][O^*] k_{ER}^{NO} + [O][N^*] k_{ER}^{ON} + [N^*][O^*] k_{LH}^{NO})}{Z_O}$$

$$Z_A = [A] \sqrt{kT/(2\pi m_A)}$$

I. Armenise, M. Barbato, M. Capitelli, and E. Kustova, "State-to-state catalytic models, kinetics, and transport in hypersonic boundary layers," Journal of Thermophysics and Heat Transfer 20, 465–476 (2006)

M. Barbato, S. Reggiani, C. Bruno, and J. Muylaert, "Model for heterogeneous catalysis on metal surfaces with applications to hypersonic flows," Journal of Thermophysics and Heat Transfer 14, 412–420 (2000)

SCIROCCO Plasma Wind Tunnel test case: Heat Flux at wall

- StS-1: recombining molecules have the same distribution of incoming ones
- StS-2: recombining molecules have uniform distributions
- StS-3: recombining molecules populate the highest vibrational level

Total heat flux at the stagnation point

	Exp.	Park (err. %)	StS-1 (err. %)	StS-2 (err. %)	StS-3 (err. %)	Park FC (err. %)
q_{probe} [kW/m ²]	1543	1708 (10.8%)	1873 (21.38%)	1816 (17.69%)	1774 (14.97%)	2160 (39.99%)

Decomposition of the total heat flux at the stagnation point

[kW/m ²]	Park	StS-1	StS-2	StS-3	Park FC
roto-translational (% contribution)	1082 (63.34%)	1213.3 (64.78%)	1516.0 (83.48%)	1763.8 (99.41%)	1146.6 (53.07%)
diffusive (% contribution)	606.10 (35.48%)	660.17 (35.25%)	300.0 (16.52%)	10.38 (0.59%)	1000.5 (46.31%)
vibrational (% contribution)	20.18 (1.18%)				13.3 (0.62%)

- A detailed description of the molecular structure is needed to predict macroscopic behaviors of high-enthalpy flows
- The StS model provides a better agreement with experiments when both thermal and chemical non-equilibrium are not negligible. This behavior is due to non-Boltzmann distributions of vibrational levels that strongly affect reaction rates
- The MiT reduces the computational cost keeping a good accuracy
- Finite rate catalysis can be useful also for metal surfaces

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